

CLASS 9th

CHEMISTRY BASIC NOTES

A. Matter & Its Nature

- Matter: Anything that has mass and occupies space.
- States of Matter: Solid, Liquid, Gas (based on shape & volume).
- Properties: Physical (color, melting point), Chemical (reaction with acids/bases).
- Mixtures & Compounds:
 - Mixture: Can be separated physically (homogeneous, heterogeneous)
 - Compound: Can be separated chemically

B. Atomic Structure

- Atom: Smallest unit of an element.
- Structure: Nucleus (protons + neutrons), Electrons in shells.
- Atomic Number (Z): Number of protons
- Mass Number (A): Protons + Neutrons
- Isotopes: Same Z, different A
- Isobars: Same A, different Z

C. Chemical Reactions

- Types:
 - Combination: $A + B \rightarrow AB$
 - Decomposition: $AB \rightarrow A + B$
 - Displacement: $A + BC \rightarrow B + AC$
 - Double Displacement: $AB + CD \rightarrow AD + CB$
 - Oxidation-Reduction: Electron transfer reactions
- Law of Conservation of Mass: Mass of reactants = Mass of products

D. Periodic Table

- Mendeleev's Table: Arranged by atomic mass
- Modern Table: Arranged by atomic number
- Groups & Periods: Vertical columns = Groups, Horizontal rows = Periods
- Valency: Number of electrons lost/gained/sharing

E. Chemical Bonding

- Ionic Bond: Electron transfer (NaCl)
- Covalent Bond: Electron sharing (H₂O, CO₂)
- Metallic Bond: Metal atoms share delocalized electrons

F. Acids, Bases & Salts

- Acid: Proton donor, sour, pH < 7
- Base: Proton acceptor, bitter, pH > 7
- Salt: Acid + Base reaction → Salt + Water (Neutralization)

G. Metals & Non-Metals

- Metals: Malleable, ductile, conductor
- Non-Metals: Brittle, poor conductor
- Reactivity series: K, Na, Ca... Au, Pt